certificateAuthorityManagement

Service Description

Service ID: *"certificateAuthorityManagement"*

Abstract
This document describes the `certificateAuthorityManagement` service, version 1.0.0, produced by the CertificateAuthority support system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface **4**
3.1 operation list-certificates . . . 4
3.2 operation revoke-certificate . . . 4
3.3 operation revoke-certificate-by-content . . . 4
3.4 operation list-keys . . . 4
3.5 operation add-key . . . 4
3.6 operation delete-key . . . 4

4 Information Model **5**
4.1 struct CertificateRecord . . . 5
4.2 struct RemovalResult . . . 5
4.3 struct RevocationRequest . . . 5
4.4 struct RevocationResult . . . 5
4.5 struct TrustedKeyRecord . . . 5
4.6 struct TrustedKeyRequest . . . 6
4.7 struct TrustedKeyResult . . . 6
4.8 Primitives . . . 6

5 References **7**

6 Revision History **8**
6.1 Amendments . . . 8
6.2 Quality Assurance . . . 8

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `certificateAuthorityManagement` service, which provides operator management of the certificate register and of trusted keys, including revocation. It is produced by the CertificateAuthority system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Certificate Authority management endpoints.

2.2 How This Service Is Meant to Be Used

Sysop lists issued certificates, revokes them by record or by content (also certificates the CA did not issue through `sign`, e.g. deployment certificates), and maintains trusted keys. Onboarding may revoke the certificates it requested.

2.3 Important Delimitations and Dependencies

Only Sysop and CertificateAuthority (Onboarding for its own records). Revocation cannot be undone; every change republishes the CRL.

3 Service Interface

This section describes the interfaces to the `certificateAuthorityManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

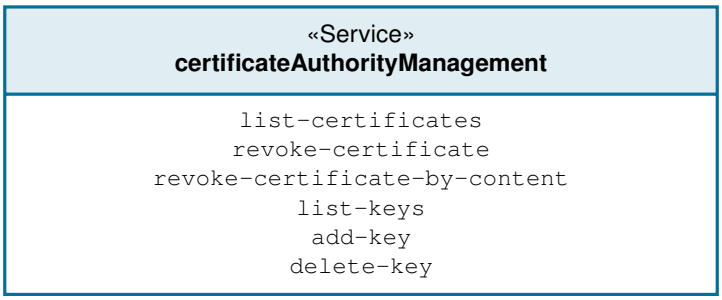


Figure 1: SysML-style block of the `certificateAuthorityManagement` service and its operations.

The following interface operations are available.

3.1 operation list-certificates () : List<CertificateRecord>

Returns the certificate register, newest first.

3.2 operation revoke-certificate (Identifier) : RemovalResult

Revokes a register record by identifier (`{id}`) and republishes the CRL.

3.3 operation revoke-certificate-by-content (RevocationRequest) : RevocationResult

Revokes any certificate of this CA given by content; unknown ones are added to the register as revoked.

3.4 operation list-keys () : List<TrustedKeyRecord>

Returns the trusted keys.

3.5 operation add-key (TrustedKeyRequest) : TrustedKeyResult

Adds a trusted key with validity.

3.6 operation delete-key (Identifier) : RemovalResult

Deletes a trusted key by identifier (`{id}`).

4 Information Model

Here, all data objects that can be part of the `certificateAuthorityManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.8, which are used to represent things like hashes and identifiers.

4.1 struct CertificateRecord

Field	Type	Description
id	Number	Record identifier.
serial_number	String	Serial number.
common_name	String	Subject common name.
created_by	String	Requesting system (`revoked-by:<caller>` for imported revocations).
valid_from	String	Start of validity.
valid_until	String	End of validity.
revoked_at	String	Revocation time, null if not revoked.
certificate_pem	String	The certificate, PEM.

4.2 struct RemovalResult

Field	Type	Description
success	Boolean	Whether a record was changed; 404 when false.

4.3 struct RevocationRequest

Field	Type	Description
certificate	String	Certificate to revoke, PEM or Base64 DER; must be issued by this cloud CA.

4.4 struct RevocationResult

Field	Type	Description
id	Number	Register record of the certificate.
serialNumber	String	Serial number.
revoked	Boolean	Always true.
changed	Boolean	False if it was already revoked.

4.5 struct TrustedKeyRecord

Field	Type	Description
id	Number	Record identifier.
public_key_pem	String	The key, PEM.
description	String	Description.
valid_after	String	Start of validity.
valid_before	String	End of validity.
created_at	String	Creation time.

4.6 struct TrustedKeyRequest

Field	Type	Description
publicKey	String	PEM SubjectPublicKeyInfo (alias `publicKeyPem`).
description	String	Description.
validAfter	DateTime	Start of validity; default now.
validBefore	DateTime	End of validity; default now plus `CERT_VALIDITY_DAYS`.

4.7 struct TrustedKeyResult

Field	Type	Description
id	Number	Record identifier.
trusted	Boolean	Always true.

4.8 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “CertificateAuthority – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-CertificateAuthority-java-spring/documentation/v5.3.0/CertificateAuthority-v5.3.0-SysD.pdf.

[2] ——, “certificateAuthorityManagement HTTP/TLS/JSON – Interface Design Description (CertificateAuthority),” Eclipse Arrowhead, IDD 5.3.0, 2026, certificateAuthority-v5.3.0-certificateAuthorityManagement-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	